

Math 2210 Quiz Review

- ① Define linear transformations & show that if S & T are linear transformations then so are $S+T$ & $S \circ T$.
- ② Find the linear transformation that sends
 - a) $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ to $\begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix}$
 - b) $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ to $\begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix}$
 - c) $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ to $\begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix}$
 - d) $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ to $\begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 2 \\ 6 \end{pmatrix}$
- ③ If T is a linear transformation that sends $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ to $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, then where does $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ go to?
 (How about $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$? How about $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$?)
- ④ Define Markov matrices and prove that $\lambda=1$ is always an eigenvalue. Which of the following is Markov?

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{pmatrix} \quad \begin{pmatrix} \frac{1}{3} & \frac{1}{6} \\ \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{6} \end{pmatrix}$$
- ⑤ Find the eigenvalues of the Markov Matrix $A = \begin{bmatrix} .90 & .15 \\ .10 & .85 \end{bmatrix}$ & the steady state eigenvector of eigenvalue $\lambda_1=1$
 (Repeat with $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$ & $A = \begin{bmatrix} .2 & 1 \\ .8 & 0 \end{bmatrix}$)
- ⑥ Show that if A & B are similar then $P_A = P_B$.
- ⑦ Show that if J is in JCF then $P_J(J) = 0$.